Omar Mohamed Adel Swellam

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EDUCATION

The American University in Cairo (AUC), Egypt, 2022 - Present
Bachelor of Science, Mechanical Engineering with a concentration in Design and Mechatronics
Minor: Economics
GPA: 3.7/4.00; expected graduation May 2027

British School in Cairo (BSC), Egypt, 2022
IGCSE

WORK AND RESEARCH EXPERIENCE

Research Assistant, Micro Linear Actuators (MLA), AUC, March 2026 - Present

- Develop compact micro linear actuators driven by shape memory alloys (SMA): design, test rigs and testing.
- Built an automated Abaqus FEA design search over 3,000+ candidate ratchet-pawl geometries for 3D-printed resin parts; found 20 designs that meet all stress and locking limits.

Intern, SLB (Reservoir Performance Services), July 2025 - August 2025

- Worked alongside engineers and technicians in wireline, well testing and coiled tubing operations at base.
- Took part in tool preparation, maintenance and calibration, and in equipment testing ahead of field jobs.

Intern, El Sewedy Electric, July 2024 - August 2024

- Learned cable and wiring manufacturing processes on the production floor.
- Supported the selection of electrical equipment and cable accessories for oil and gas projects against industry standards.

Intern, Raya Holdings, January 2024 - February 2024

- Helped set up Raya's Leadership Needs Analysis (LNA) program: interviewed employees on their training needs and career goals.
- Negotiated contracts with training providers to get the most value for the budget.
- Co-organized two large company events with several hundred attendees each.

Intern, Philips, August 2023 - September 2023

- Worked on the first mobile MRI truck in Africa and assisted in installing an MRI system.
- Carried out scheduled and unscheduled maintenance on MRI, CT, X-ray and IGT machines.

ACADEMIC PROJECTS

Five-Axis 3D Printer, Senior Thesis, AUC, 2026 - 2027 (Thesis Coordinator, team of 5)

- **Leadership:** since September 2026, assign tasks, keep the team's progress log, set weekly priorities and keep five members motivated and on schedule.
- Built slicing software that curves print layers along the load paths of a part (for strength) or around its overhangs (for less support), using finite element analysis, collision checks and CuraEngine toolpaths.
- Designed parts of the machine CAD, including the tilting bed's gimbal joint, and wrote the printer's custom five-axis firmware kinematics.
- Used agentic AI to build a simplified mass-spring-damper model of the printer that tests its vibration under different motions in seconds; used its predicted loads to check the critical brackets in FEA.
- On a 40-part benchmark vs normal flat slicing: support material 32.1 to 21.0 g (-35%; none at all on 14 parts), overhang area needing support -57%, and zero collisions.
- Strength mode: stress pulling across the layers (the weak direction) cut on 7 of 8 test parts, by up to 56%.

Finite Element Methods and Applications in Design, MENG 4553, AUC, Fall 2026

- Final project: full Abaqus simulation pipeline for a manufactured notched shaft, from thermal and manufacturing effects through plasticity and damage to the final stress state at the notch.

- Take-home: built a three-tier submodeling chain to resolve a sharp notch within a 1,000-node license limit, and verified mesh convergence with ASME V&V 10 Richardson extrapolation.
- Labs in heat transfer, pin-in-lug contact and thermo-elastic-plastic analysis; automated the post-processing with Python and used AI to generate the HTML result reports.

Modeling and Control of a Quadcopter, System Dynamics MENG 3705, AUC, Fall 2025 (Project Coordinator, team of 5)

- Derived a nonlinear 6-DOF model, linearized it about hover and built the MATLAB/Simulink simulation.
- Designed a cascaded PID controller (position outer loop, attitude inner loop): every axis tracks a unit step with under 10% overshoot, attitude settles in about 3 s, and the drone follows a 3D spiral trajectory.
- Ran sensitivity studies on thrust-to-weight ratio and center-of-gravity position.

Smart Tennis Ball Launcher, Electromechanical Systems MENG 4779, AUC, Spring 2026 (Project Coordinator, team of 5)

- Led the mechanical hardware: a counter-rotating two-wheel launcher designed in SolidWorks and 3D printed.
- Wrote PLC ladder logic (start/stop, timed auto-shutdown, ball counter) and verified it in RSLogix.

Policy Brief: Beyond Fiscal Success, the Limits of Subsidy Reform in Morocco, ECON 4051 Seminar on Economic Development in the Middle East, AUC (team of 4)

- Evaluated Morocco's conditional cash transfer program (2.3 million children, 434 municipalities): school dropout fell 1.3 pp (1.8 pp for girls) and secondary enrolment rose 4.5 pp.
- Argued that cash transfers are necessary but not sufficient: poverty rose back to 4.9% in 2022, youth unemployment is 35.8%, and female labor participation is 19% vs 69% for men, a gap costing an estimated 7-12% of GDP.

EXTRACURRICULAR ACTIVITIES

Mechanical Team Lead, AUC Aquavaders (MATE ROV Competition), June 2026 - Present

- Lead the mechanical team of an underwater robot (ROV): delegate tasks and manage timelines.
- Direct propulsion, buoyancy and structural design, CAD modeling, CFD and FEA simulation.
- Coordinate with the software and electronics teams to integrate mechanical, electrical and control systems; oversee fabrication, 3D printing and testing.

Mechanical Team Member, AUC Aquavaders, June 2025 - May 2026

- Designed propulsion and buoyancy systems (component placement, stability) and ran CFD to improve hydrodynamic performance.

COMPUTER AND LINGUISTIC ABILITIES

- **Design & CAD:** SolidWorks, Fusion 360, AutoCAD.
- **Simulation & Analysis:** FEA and CFD (ANSYS), Abaqus; surrogate modeling.
- **Agentic AI:** run teams of AI coding agents (Claude Code) with a shared task board, review and test their work; AI-built engineering tools and automated reports.
- **Programming:** Python, MATLAB/Simulink, Git.
- **Controls & Systems:** PLC-based control systems (ladder logic), pneumatic systems.
- **Manufacturing:** CAM/CNC, 3D printing and additive manufacturing.
- **Software:** Microsoft Excel, PowerPoint and Word; GitHub; Ultimaker Cura; RSLogix.
- **Languages:** Fluent in Arabic and English.